

Design and development of intensive aquaculture supervising model

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ABSTRACT

Aquaculture is a flourishing business sector, from many decades. India holds the second place in the international level for fresh water aquaculture cultivation. Despite of that a large extent of Indian aquaculture is as yet dependent on customary cultivating techniques. Changing over them to current cultivating techniques will build fish production and will also address sustainability of environment. This research concentrates on supervising essential water parameters like pH, turbidity and temperature, which is essential to monitor and control these parameters to enhance the fish growth and their production. To make the system more robust and agile, Raspberry pi 3B+ is utilized and it has inbuilt Wi-Fi module so that sensor data can be uploaded to cloud for worldwide access. The proposed system consists of microprocessor, sensors and web application for data acquisition and to supervise essential water parameters. Water heating device and water pump is used as correcting devices to keep threshold level of these water parameters in standard to enhance the fish growth. To check the system reliability and efficiency, two aquaculture setups were made, one is conventional method and another is supervising method. The data acquired from supervised method was 7.5 to 8.5 pH, less than 250 NTU turbidity and temperature were between 26 to 29°C, which helped in lowering mortality rate of fishes and also lowered the work and time consumption by aqua farmers.

1. Introduction

Most of us think that, the fish we consume originates from the ocean, streams and lakes. In fact, it is not true, actually it is from fish farms. Nearly about 1.07% Gross Domestic Product (GDP) is from aquaculture, which is one of the booming sectors in India [1]. As we are witnessing from past couple of years, fish and fish products in this country is becoming one of the largest sectors in exporting, nearly 50 different kinds of fish and fish products is exported to 75 countries around the world. In recent census it was mentioned that fish and fish products requirement may go up due exploitation of natural resource fishing. To overcome these problems and to take advantage of the situation, i.e., increase in demand for aqua-products, farmers started to do commercial aquaculture. Because it is a new terrain to explore, farmers are facing difficulty in culturing fishes due to unpredictable climatic conditions [2]. At present farmers are following unconventional manual ways of water parameter testing methods which will consume lot of time. Even though testing is done, this may not be accurate because water quality changes with time as they are not tested frequently [3]. India has to address the problems associated in developing the fish cultivating infrastructures, including the following technical challenges. First, absence of other potential aqua feed-devouring species or high esteem fish in India, and

this ought to be a thought for new presentations. For instance, tilapia cultivating is yet to take off in the nation. Second, freshwater fish cultivating is as yet dependent on conventional strategies, huge lakes, no water trade, no depleting, and no base residue expulsion that regularly lead to conditions that advance sickness. In order to address these issues, latest technology should be applied in aquaculture farming which may improve the production and reduce the loss by constantly supervising the quality of water parameters [4]. In this research, we are supervising the essential water parameters like pH, turbidity and temperature in real time and same prototype model is developed which is compact in size, low cost and plug and play.

2. Related Works

From the literature survey it was observed that, several authors have investigated and conducted several experiments to curtail the impact of water quality in fish growth by using information and communication technologies (ICT) and made an attempt to resolve the issues [5]. In the literature [6], authors have used the pH, dissolved oxygen (DO) and turbidity sensors to determine the few of the water parameters and were able to provide solutions to determine the water parameter using non-invasive method. The ideal fish production will totally dependents upon many compound, physical and organic qualities of water which

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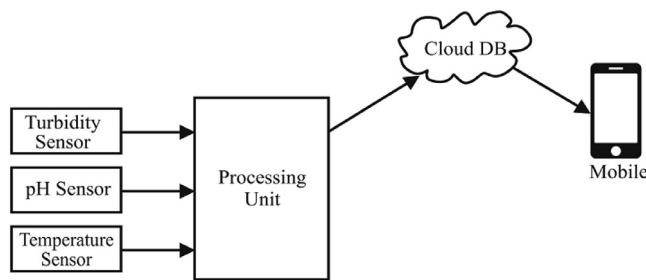


Fig. 1. System architecture.

will have greater impact. Thus, compelling the lake board authority requires an acknowledgment of water standards. Water standards is obtained by factors like turbidity, temperature, pH, straight forwardness, alkalinity, water tone, carbon dioxide, conductivity, saltiness, hardness, DO, total dissolved solids (TDS), unionized smelling salts, nitrate, essential efficiency, microscopic fish population, Biochemical oxygen demand (BOD) and many more [7]. In the literature [8-9], the authors have used the Arduino device as miniature microcontroller for checking the water life, but Raspberry Pi-3 has inbuilt Wi-Fi module when compared with Arduino. Author Raghu Sita Rama et al, has conducted experiments for supervising aquaculture parameters using real time monitoring with internet of things (IoT) devices. They have used Nitrate, Salt, Ammonia, Temperature and pH sensors for monitoring the water quality [10]. But incorporating many sensors may increase the cost of the project. Maxime Lafont et al has used Temperature, DO, Lora for communication and pH sensors for monitoring the water parameters for aquaculture needs in real time with the title “Back to the future”. But this paper mainly concentrates on 5G technology which is not yet available in many nations and total project cost may also increase because of using Lora device for communication [11]. In literature [12-13] authors have used nodeMCu microcontroller to monitor the water quality parameter using sensors like dissolved oxygen, temperature, pH, and color sensor. These sensor data are used to detect color of the water, based on this information decision are made whether color is present or not in the water. In literature [14-17] authors achieved water quality monitoring by detecting pH level, temperature, turbidity, salinity, dissolved oxygen of the water. Also they have used Arduino mega for acquiring data from sensor, raspberry pi, loRaWAN for communication between sensor and user interface web application. In literature [18-20] authors have used big data technologies and neural network model to predict the change of water quality parameters, monitoring abnormal phenomena, prevent the occurrence of diseases in time, and reduce the cost of breeding [21-25].

3. Methodology

3.1. System architecture

In this research work, we are mainly concentrating on supervising essential water parameters of the aqua life and to alert if standard level is not met. The essential water parameters like pH, turbidity and temperature are considered as important parameters to monitor. Fig. 1 shows the system architecture of the model developed, which includes processing unit and water parameter sensors such as Temperature sensor (DS18B20), Turbidity sensor, and pH sensor. The above mentioned sensors are linked to Raspberry Pi 3B+. The standard values for the water parameters are pre-programmed in the processor to test whether the data acquired satisfy the desired values. When the range aren't met, alert message will be sent to farmer. The acquired sensor data are sent to the cloud via Wi-Fi IoT Protocol and displayed in the Web application. Also, ThingSpeak cloud is used to store the sensor data, an IoT analytics platform service will allow to aggregate, visualize, and analyze live data streams in the cloud. The data to ThingSpeak can be sent using HTTP

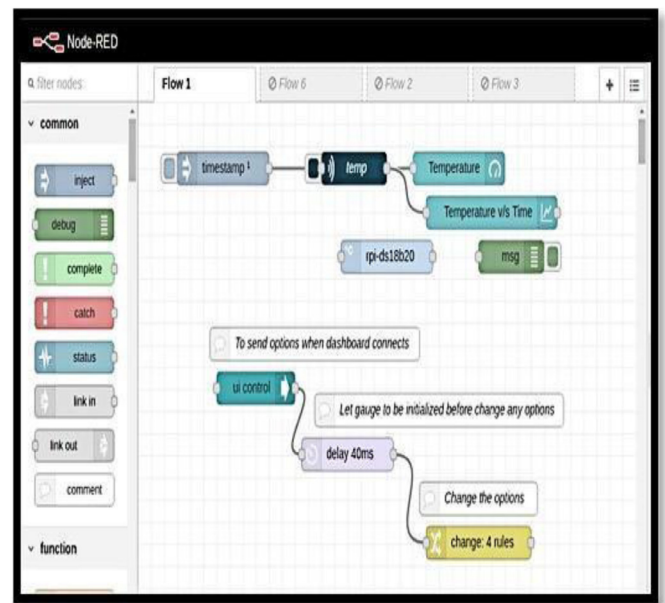


Fig. 2. Node-red development tool.

protocol. This will create instant visualization of live data and will send alerts.

The relationship between turbidity value and voltage from sensor is given by

$$Y = -1120X^2 + 5742X - 4352.9 \quad (1)$$

Where:

Y = turbidity value in NTU (Nephelometric Turbidity unit) 1NTU = 1 mg/L

X = voltage

The relationship between pH value and voltage from sensor is given by

$$E = E^\circ + \left(\frac{R \cdot T}{n \cdot F} \right) * 2.303 * pH \quad (2)$$

Where:

E = total electrical potential difference in millivolts (mV)

E° = electrical potential standard produced by the reference electrode (mV) which is constant

R = universal gas constant (8.3145 J/mol.K) n = charge/valency of the ion (for hydrogen ion n = 1)

F = Faraday constant (96485.309 C/mol)

At a temperature 25°C (298.15 K) the above equation becomes

$$E = E^\circ + \left(\frac{8.3145 * 298.16}{96485.309} \right) * 2.303 * pH \quad (3)$$

$$E = E^\circ + (0.059 * pH) \quad (4)$$

3.2. Web application

The web application is a user interface (UI) page which is displayed to end user, so that the user can search this page on browser and can monitor this essential water parameter conditions in the fish tank. To build web application Node-Red was used, it is a flow-based developing tool for visual programming as shown in Fig. 2 and for programming JavaScript and Cascading Style Scripts (CSS) were employed.

The Node-Red web application result is shown in Fig. 3. This is used to gather all the sensor data and display the real time sensor results. For sensor data acquisition and to analyze the acquired data if it's crossing threshold level or not and to send the data to cloud, Python scripting is used, for web application to develop UI JavaScript is used.

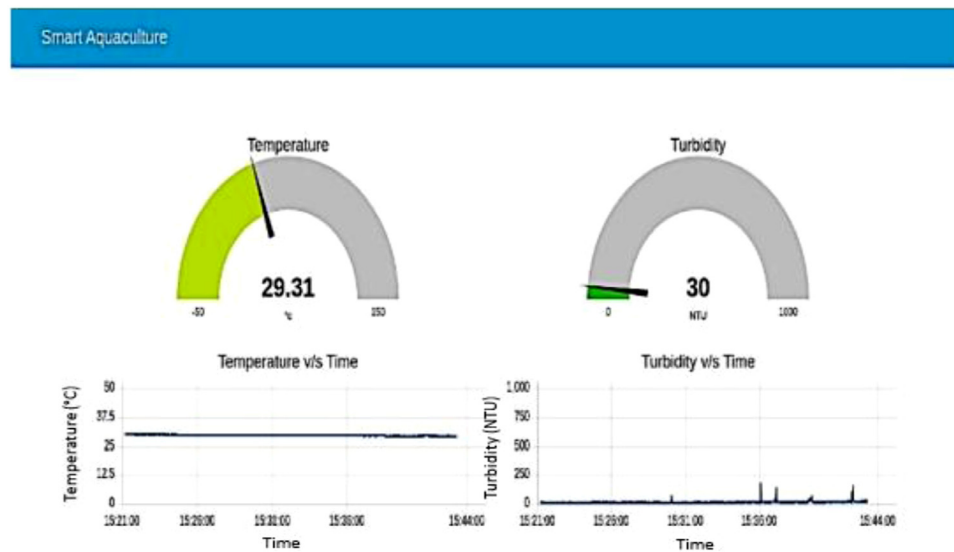


Fig. 3. Web application.

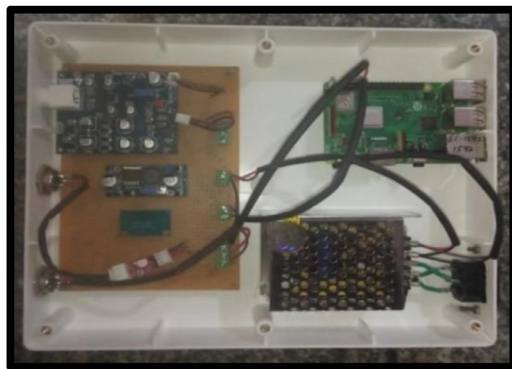


Fig. 4. Hardware design.

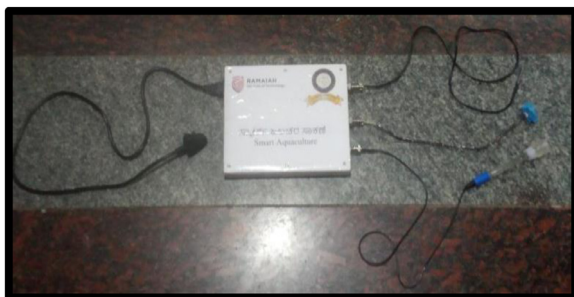


Fig. 5. Plug and play water supervising model.

4. Results

In the literature reported [14–17], water monitoring system makes use of two controlling units, one is Arduino for acquiring sensors data and another is Raspberry-pi for establishing communication between user interface and they have incorporated many sensors which increases the overall cost of the project. The Fig. 4 shows the hardware model developed and Fig. 5 shows the sensor integrated to the hardware for supervising the water parameters.

The real time monitoring temperature sensor represented in degree Celsius, reading gauge and temperature vs time graph is shown in Fig.s 6, where x axis denotes time in HH:MM:SS format and y axis represents temperature in degree Celsius. Ideal water temperature for

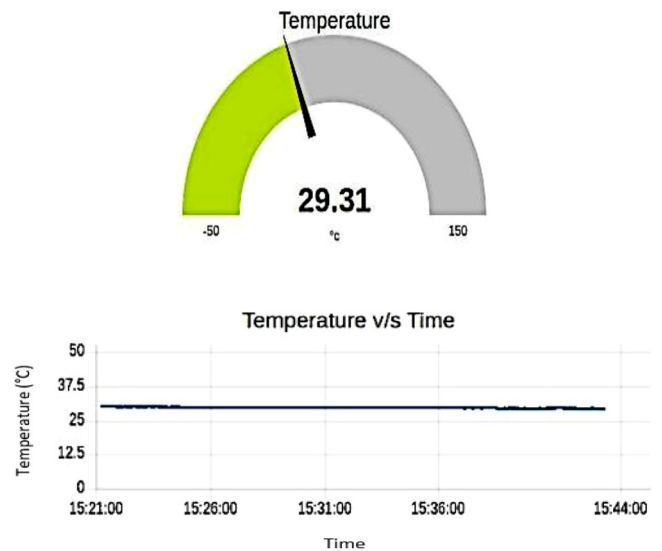


Fig. 6. Real time monitoring of temperature.

fish ranges from 25–27°C. If this range exceeds, an alert is sent to user. If the temperature is reduced, which is less than the standard range, then to prevent temperature dropping from ideal condition water heating device is turned on.

The real time monitoring turbidity sensor represented in NTU (Nephelometric Turbidity Unit), reading gauge and turbidity vs time graph is as shown in Fig.s 7, where x axis denotes time in HH:MM:SS format and y axis indicates turbidity in NTU. Ideal water turbidity for fish is less than 250 NTU. If this range exceeds, an alert is sent to aqua farmer. So that the user can change the water in fish tank.

The real time monitoring pH sensor, reading gauge and turbidity vs time graph is as shown in Fig.s 8, where x axis indicates time in

Table 1
Water parameters threshold values.

Sl.No.	Parameters	Range
1.	Turbidity	<250NTU
2.	pH	7.5–8.5
3.	Temperature	26–29 °C

Table 2
Comparison of sensors and software tools.

Sl. No.	Particulars	Components	Software tools
1.	Proposed model	Raspberry Pi, pH sensor, turbidity sensor, temperature sensor	Node-RED, Node.js, python, ThingSpeak cloud
2.	Ferdin Joe John Joseph (2020)	Arduino Uno, pH sensor, DHT 11, CO sensor, DS18B20, LDR sensor	Arduino IDE, Jupyter Notebook
3.	Lean Karlo S. Tolentino (2020)	Arduino Mega, Raspberry Pi, LoRaWAN, temperature sensor, pH sensor, ORP sensor, turbidity sensor, salinity sensor, dissolved oxygen sensor	Arduino IDE, HTML, CSS and JavaScript
4.	R. Hafid Hardyanto (2019)	Arduino Mega, ESP8266 module, humidity sensor, water level sensor, light sensor	Arduino IDE
5.	J.Zoldeo (2019)	Raspberry Pi, pH sensor, water level sensor, temperature sensor	Arduino IDE, HTML, SCSS, TYPE SCRIPT

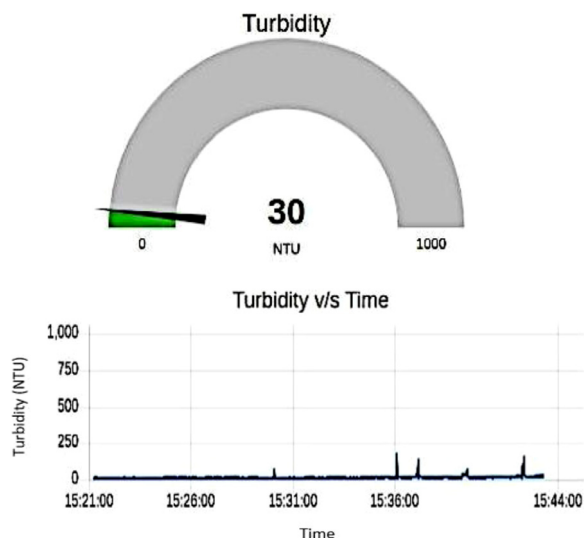


Fig. 7. Real time monitoring of turbidity.

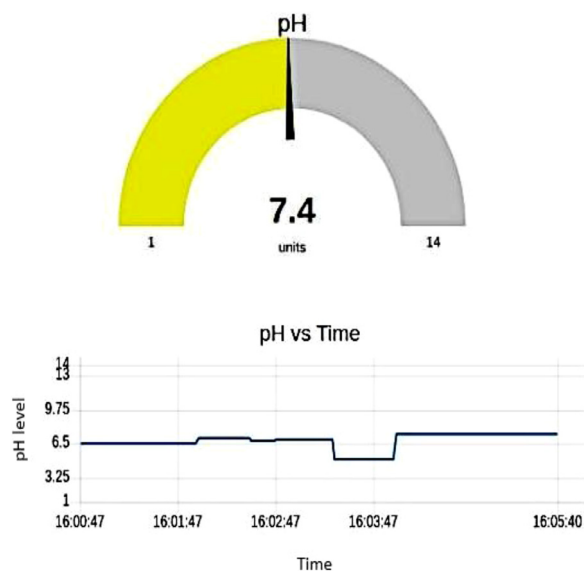


Fig. 8. Real time monitoring of pH level.

HH:MM:SS format and y axis denotes pH level from 0 to 14. The ideal water turbidity for fish family ranges from 6 to 9 (unit less). If this range exceeds, an alert is sent to aqua farmer. So that they can take corrective measures to improve the level of pH in fish tank.

The standard threshold levels of turbidity, pH and temperature for sustainable growth of fish family was suggested by research professor at Fisheries Research and Information Centre (Inland), KVAFSU, Bangalore and same is shown in [table 1](#).

The [table 2](#) shows the comparison of sensors, processors and software tools used in the cited reference papers and the proposed model. As discussed in related work, for monitoring the water quality authors have employed both Raspberry Pi and Arduino devices, which increases the cost of the system. Apart from that they have integrated several sensing devices. Combining such a large number of water monitoring sensors in a single microcontroller may cause system slow. Also, most of them have not used cloud storage concept, but in the proposed system ThingSpeak cloud is utilized which is also a cost effective and helps in storing sensor data for remote access.

5. Conclusion

The proposed system is developed by incorporating information and communication technology. In addition, the proposed model will monitor temperature, turbidity and pH in the aqua ponds using low cost reliable sensors. The projected system is compact and can be used as plug and play device. All the above mentioned sensors are interfaced with Raspberry-pi which has inbuilt Wi-Fi module, so that sensor data can be uploaded to cloud for research purpose and it can be accessed from remote areas. This system also gives alerts whenever water quality parameter crosses the threshold value. Thus, distant administration of aquaculture in rural areas can be achieved effectively.

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